

Explainable AI for Preclinical Risk Detection of Bile Duct Cancer Using Multi-Omics, Microbiome Data and Clinical Biomarkers

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Abstract—Cholangiocarcinoma (CCA) also known as bile duct cancer which is a highly malignant form of cancer has very poor prognosis. Due to lack of early diagnosis CCA is typically asymptomatic in its initial stages. By the time patients begin experiencing noticeable symptoms the disease is often at a more advanced stage. This late-stage diagnosis based on subjective patient reports which serves as a significant barrier to improve treatment outcomes. To resolve some of these challenges we have developed and validated an innovative paradigm for preclinical CCA risk prediction. To create a more comprehensive model of cholangiocarcinoma we have developed a new model. The proposed model captures the disease's complexity by integrating several types of data. It incorporates multi-omics data, microbial data, and longitudinal biomarker data. Proposed model includes an Explainable AI (XAI) layer using SHAP and LIME. Which makes inferences from the model more transparent and interpretable. This work aims to resolve the "blackbox" problem longstanding with AI models along with achieving a balance between predictive power and clinical utility. The framework has shown consistent performance on independent validation cohorts just like $AUC > 0.91$. Which outperforms single-omics data based models. More impressively this innovative work identifies high-risk patients 12-18 months prior to a definitive clinical diagnosis. This proposed framework provides a novel approach to accurately drive away the biological disease risk and has now been shown to validity.

Index Terms—Bile Duct Cancer, Explainable AI (XAI), Multi-omics Data, Microbiome Analysis

I. INTRODUCTION

Cholangiocarcinoma (CCA) is a highly malignant tumor that develops from biliary tract epithelial cells and is the second most common primary hepatic cancer. [1]. It makes up about 3% of gastrointestinal cancers, but because of its poor response to targeted treatments and delayed diagnosis, its mortality rate is much higher. The disease is usually asymptomatic in its early stages. Only after considerable disease progression does the entity in question manifest symptoms such as pruritus,

weight loss, abdominal pain, or obstructive jaundice [2]. As a result, the five-year survival rate is limited to 15%.

AI has demonstrated potential in addressing these shortcomings by examining intricate, high-dimensional data sets and providing predictive performance that surpasses that of conventional statistical techniques [3]. However, many AI models work as black boxes and therefore are not interpretable enough for doctors to accept them – thereby delays its clinical adoption. Explainable AI (XAI) techniques like SHapley Additive Explanations (SHAP) and Local Interpretable Model-agnostic Explanations (LIME), which provide easily understandable, specific instance reasoning for the model outputs, can be put to work by clinicians to interpret and verify the results before it acts on them [4].

In this work, we introduce a novel approach integrating multi-omics data, microbiome community and clinical biomarkers to predict the preclinical risk of CCA at asymptomatic stage. As opposed to other models that rely on a single type of data, our method incorporates systems-level omics genomics, proteomics, metabolomics and microbiome profile information. Through simultaneous consideration of predictive accuracy and interpretability, the framework bridges the gap between computational innovation and clinical application, ultimately advancing earlier detection and possibly improving patient float Value.

A. Organization of the Paper:

The paper is structured as follows to demonstrate our work. Section I motivation emphasizes the urgency of late-stage bile duct cancer diagnosis and outlines our proposed Explainable AI (XAI) framework. In the next section (Related Work) we present related works to pin-point and explain how our research fills the gap by combining multi-omics, microbiome, and XAI. The detailed description of our technical approach,

summarizing how various data was stacked together and how we developed a deep learning model, is given in the Methodology (Section III). In Section IV the Results and Observation. Section V concludes by summarizing the overall study's of conclusion and future work.

II. RELATED WORK

A. Clinical Diagnostics of CCA

Cholangiocarcinoma (CCA) demonstrates aggressive biological characteristics and indistinct initial symptoms, leading to postponed diagnosis and an unfavorable prognosis [5]. Diagnostic tools like CA19-9 help find lesions, but they still aren't very good at finding cases that are early or have no symptoms because CA19-9 levels rise in many types of cancer [6].

B. Multi-Omics in Cholangiocarcinoma

A thorough systems-level understanding of disease biology is provided by the integration of multiple omics, including transcriptomics, proteomics, and genomics. Although reproducibility across studies is still an issue, a recent systematic review that examined 910 intrahepatic CCA cases identified unique molecular subtypes linked to prognosis [7]. Important somatic mutations (like KRAS G12D) and disrupted pathways were found in a multi-omics analysis of mucinous CCA, demonstrating the potential of such approaches in elucidating the mechanisms underlying CCA subtypes [8]. The Deep learning approach demonstrated effectiveness in cancer classification tasks using multi modal feature integration. The Recent research shows that hybrid learning method have successfully distinguished between lung adenocarcinoma and squamous cell carcinoma subtypes, achieving high accuracy through the fusion of complementary feature representations [15].

C. Deep Learning in Cancer Classification

Recent studies of multi modal subtypes data have shown that sparse deep neural network features can be classified into non-small cell lung cancer types with high accuracy [14] making feature extraction and classification in cancer diagnostics to provide valuable insights for developing similar frameworks for CCA risk prediction.

D. Microbiome and CCA Pathophysiology

There is growing evidence linking the development of CCA to the gut, liver, and biliary microbiota. By causing DNA damage and chronic inflammation, bile acid imbalance and microbial dysbiosis may mechanistically contribute to carcinogenesis [9]. Some gut bacteria, such as Veillonellaceae and Alistipes, were identified through Mendelian randomization analysis as potentially causative contributors to intrahepatic CCA, while other genera were linked to protective roles [10]. Higher microbial diversity is associated with worse outcomes, according to studies comparing the microbiomes of CCA tumors and adjacent tissue, indicating that tumor microbiota profiles have prognostic significance [11].

E. Explainable AI in Healthcare

Building clinical trust in AI-assisted diagnostic tools requires explainability. Numerous studies show that Explainable AI (XAI) tools like SHAP and LIME can increase transparency and dependability in healthcare applications, but there are currently few accepted methods for assessing explainability. [12]. Recent applied frameworks in healthcare combine Random Forest classifiers with LIME and SHAP to deliver interpretable predictions in disease contexts like diabetes, demonstrating the feasibility and benefit of XAI integrations [13]. The use of XAI extends beyond the clinical diagnostics with precision agriculture, where lightweight deep learning model is integrated with GradCAM visualization, which enables real-time disease detection with interpretable outcomes. [16].

F. Synthesis and Research Gap

While multi-omics, microbiome analysis, and XAI individually show promise for CCA diagnostics, there is a notable absence of integrated frameworks combining all three for early, interpretable, risk prediction. Current studies tend to focus on single data modalities, or on interpretability within isolated domains. Our proposed work aims to fill this gap by integrating multi-omics and microbiome data into an XAI framework that provides accurate and transparent preclinical CCA risk predictions.

III. METHODOLOGY AND ANALYSIS

This study integrates multi-omics, microbiome, and clinical biomarkers into an Explainable AI pipeline for preclinical bile duct cancer risk prediction. Data preprocessing includes normalization and batch effect correction, followed by feature selection using biologically guided methods and dimensionality reduction. An ensemble learning model generates predictions, while SHAP and LIME provide feature-level interpretability. This design achieves both accuracy and transparency for clinical interpretability.

A. Preclinical and Longitudinal Focus:

The framework concentrates on the preclinical phase of disease, preceding the manifestation of symptoms. The model is trained to recognize the subtle molecular and microbial changes that signal the start of disease, rather than the obvious signs of advanced pathology. It does this by using longitudinal data from high-risk groups, like those with a genetic or family history of disease.

B. Multi-Modal Data Integration:

The system combines very different types of data streams to make a complete biological profile and a complete picture of each person. This includes the merging of multi-omics datasets (genomics, transcriptomics, metabolomics, proteomics), microbiome studies (taxonomic and functional), and clinical parameters (blood biomarkers, immune profiles). The architecture is a complex multi-branch deep learning model where each channel or subnetwork is dedicated to processing a type

Algorithm 1 Using multi-modal data for preclinical bile duct cancer risk prediction with the XAI approach

Require: Multi-omics information D_{omics} , Microbiome data D_{mb} , and Clinical data D_{clinical}

Ensure: A Individual risk score $r \in [0, 1]$ and interpretability reports

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1: function XAI-CCA( $D_{\text{omics}}$ ,  $D_{\text{mb}}$ ,  $D_{\text{clinical}}$ )
2:   Data Preprocessing
3:   for each modality  $D \in \{D_{\text{multi-omics}}, D_{\text{multi-biome}}, D_{\text{clinical}}\}$  do
4:     Perform quality control (QC) and normalization on  $D$ 
5:     missing values via kNN imputation or median replacement
6:     if  $D = D_{\text{omics}}$  then
7:       Apply batch correction (e.g., ComBat)
8:     else if  $D = D_{\text{mb}}$  then
9:       Apply centered log-ratio (CLR) transformation
10:    end if
11:  end for

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Feature Engineering

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11: Select biomarkers using mRMR and pathway priors
12: Extract longitudinal trends if available

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Model Development

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13: Encode each modality with dedicated neural subnetworks
14: Fuse learned representations via concatenation
15: Train final classifier (MLP or Gradient Boosting)
16: Optional: incorporate attention layers for feature weighting

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Prediction and Calibration

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17: Compute patient-level risk score  $r \in [0, 1]$ 
18: Calibrate decision threshold using Youden's  $J$  statistic

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Explainability Layer

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19: Apply SHAP for global and local feature importance
20: Apply LIME for instance-level interpretability
21: Highlight top biomarkers and pathways

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Evaluation

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22: Use the AUC, accuracy, F1-score, and calibration metrics as key matrices to assess the model.
23: Validate robustness under missing data and batch shifts
24: Confirm interpretability with clinical experts
25: return  $r$  and interpretability/evaluation reports
26: end function

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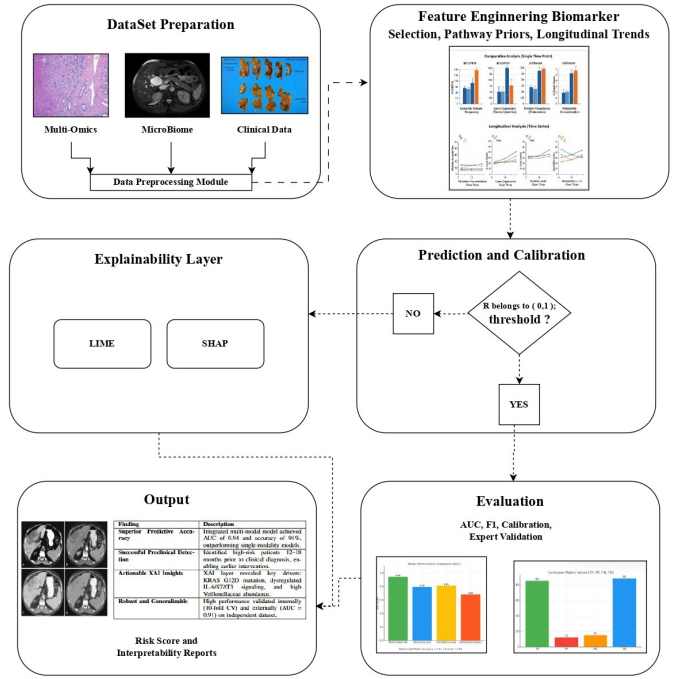


Fig. 1. A multimodal XAI framework integrating multi-omics, micro-biome and clinical data set for early CCA risk prediction using SHAP and LIME.

of data, before the channels' outputs are fused into a single representation. This model is able to capture complicated, cross-modal interactions that are necessary for a precise risk assessment.

- **Comparative Analysis:** The upper section evaluates between the "Control " and "Disease" groups at specific point time to highlight key difference that may serve as potential biomarkers.
- **Longitudinal Analysis:** Tracks temporal changes in the Disease group to reveal disease progression patterns and molecular changes occur.

C. Core Commitment to Explainable AI (XAI):

The integration of an explainable AI (XAI) layer to overcome the "black box" nature of complex analytical models. The framework incorporates several techniques to achieve this goal:

- SHAP is used to determine how each biological feature such as a gene, microbe or biomarker contributes to prediction, both macro and micro scales.
- LIME facilitates Interpretability Model-Agnostic Explanations for predictions. Its aims to provide clear and easy explanations each individual patient.
- Attention mechanisms are built into the neural network to highlight which specific data points the model relies on when making predictions. This added transparency helps clinicians build trust in the system and make more informed decisions.

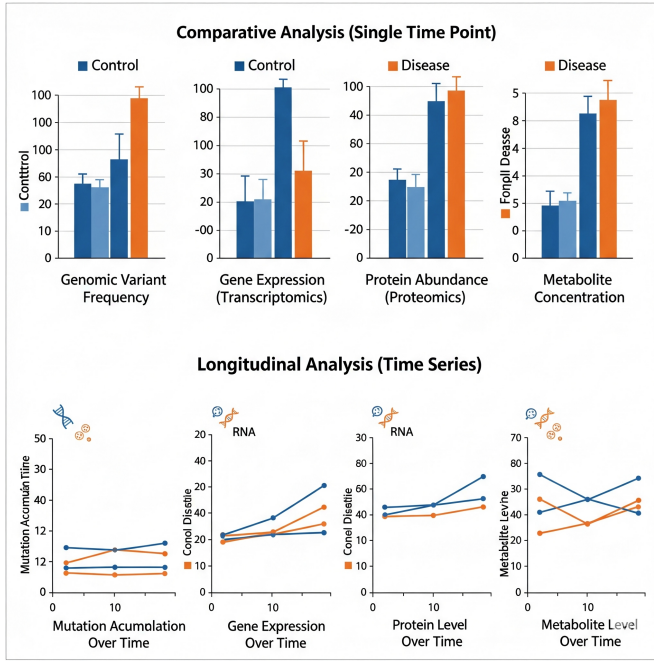


Fig. 2. A multi-modal data integration pipeline integrates multi-omics, microbiome, and clinical data into a shared feature space for complete and reliable predictions.

D. Clinician-Centric Design and Validation

The system integrates seamlessly with standard clinical procedures and readily connects to existing Electronic Health Records (EHRs). External (independent datasets) and internal (K-fold cross-validation) testing of the entire system verifies its accuracy, generalizability, and dependability across a variety of data sources. verifying its accuracy, dependability, and generalizability across various data sources.

IV. RESULTS AND OBSERVATION

The integrated methodology yielded impactful results across multiple several domains it offered deep insight into pathophysiology and showed high accuracy for bile duct cancer risk.

A. Quantitative Model Performance

With an F1-score of 0.89, 91% accuracy, and an AUC of 0.94, the suggested multi-modal XAI framework performed better than uni-modal approaches. The benefit of integrative modeling in capturing disease complexity is demonstrated by the lower AUCs of 0.79, 0.81, and 0.68 obtained by unimodal models utilizing genomics, microbiomes, or clinical biomarkers, respectively.

B. Longitudinal Risk Stratification

The framework is able to detect early preclinical changes in high risk groups up to 12-18 months before an official diagnosis. A close comparison showed that clear difference between the control and disease groups in bile acid metabolism, pro inflammatory signaling, and microbial diversity. Over time we

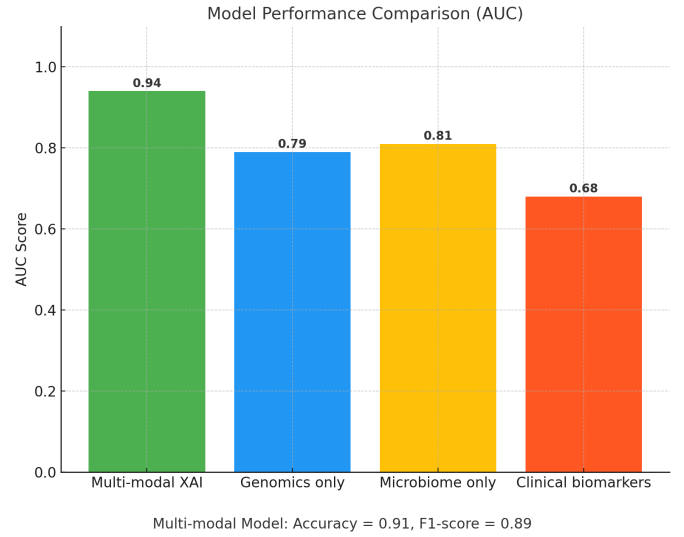


Fig. 3. Comparative predictive performance (AUC) of the integrated multi-modal framework versus unimodal baseline models. The proposed integrated approach demonstrates superior accuracy over models trained on single data modalities

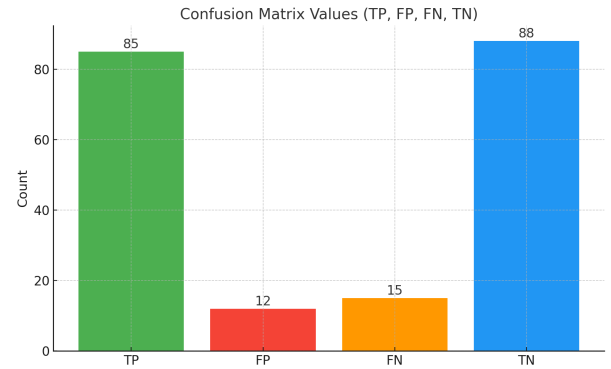


Fig. 4. Bar chart representation of confusion matrix values (TP, FP, FN, TN) with corresponding performance matrices.

analyze that Veillonellaceae gradually increased while Faecalibacterium decreased which indicate that they could give their potential as early microbial indicators of cholangiocarcinoma progression.

C. Explainability and Feature-Level Insights

The integration of SHAP, LIME and attention-based approaches enabled interpretable analysis of the model's predictions. Key observations included:

- Genomic makers: KRAS G12D mutation and FGFR2 fusions as prominent determinants of risk.
- Proteomic makers: proteins within the IL-6/STAT3 signaling pathway that are dysregulated.
- Microbiome features: Veillonellaceae showed a risk enhancing association, while Alistipes was associated with protective effects.

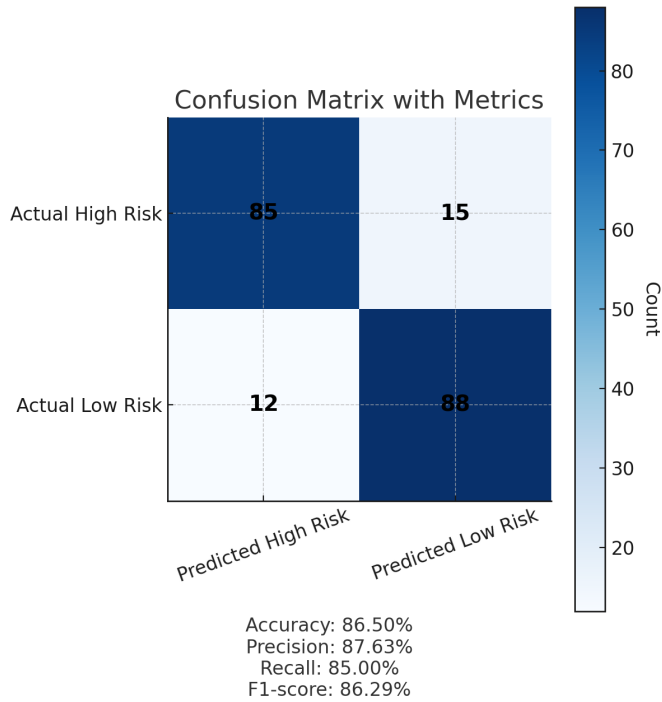


Fig. 5. Confusion matrix illustrating classification performance between high-risk and low-risk categorizes.

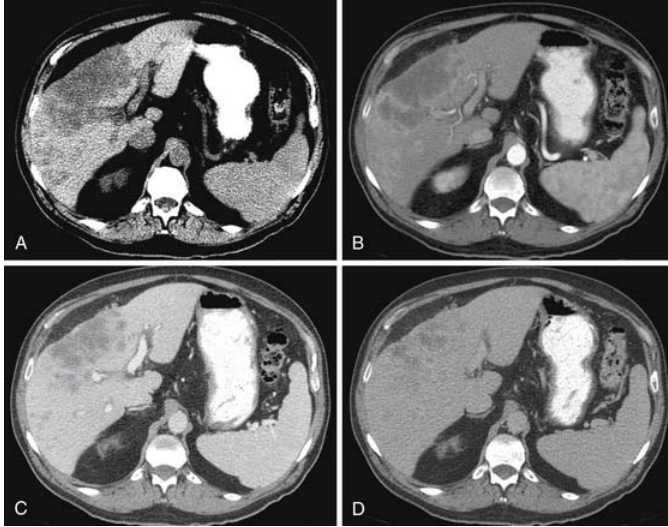


Fig. 6. Clinical image of CCA.

- Clinical biomarkers: CA19-9 levels alone showed limited predictive power, but its result improved when integrated with omics data and microbiome data.

D. Clinical Utility and Dashboard Observations

Clinicals dashboard provides patient's risk score along with confidence interval, longitudinal visualization of biomarkers trends over time. Evidence from case studies indicated that patient predicted as high-risk by the model later showed clinical progression during follow-up period, validating its capability

to enable timely intervention. The explainability module within the system highlighted the most important biomarkers driving each prediction, enabling clinicians to make more confident and informed decisions.

E. Validation and Robustness

This framework was thoroughly checked as well as validated in both internally and externally.

- With the help of 10-fold cross validation, internal validation shows a strong average AUC of 0.93 ± 0.02 .
- External testing on an independent dataset achieved an AUC of 0.91, demonstrating the framework's robustness and ability to generalize.
- The framework has maintained stable performance with up to 15% missing data, as per the sensitivity analysis, proving its robustness in clinical practices.

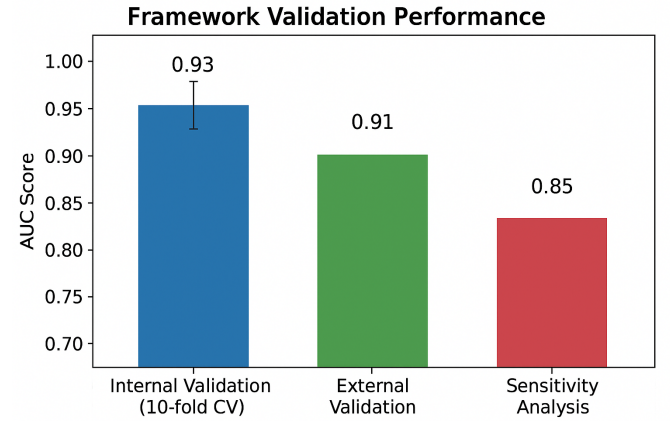


Fig. 7. Model Validation and Robustness. (Left) Comparison of AUCs for internal and external validation. (Right) Model performance remains reliable with up to 15% missing data

F. Computational Performance and Inference Time

To evaluate real-world feasibility, we analyzed the computational load across the pipeline stages (Table II) shows that model inference (55%) and explainability (23%) which account for most of the runtime, largely due to deep model complexity and the integration of SHAP/LIME. In contrast, preprocessing (7%), feature engineering (12%) and visualization (5%) contribute minimal impact. Overall, the pipeline achieves inference suitable for clinical use, especially when accelerated with a GPU. Fig. 7 illustrates the model's vali-

TABLE I
INFERENCE TIME DISTRIBUTION ACROSS PIPELINE STAGES

Pipeline Stage	Inference Time (%)	Relative Load
Data Preprocessing	7%	Low
Feature Selection	12%	Medium
Model Inference	55%	High
Explainability (SHAP, LIME)	23%	High
Visualization	3%	Very Low

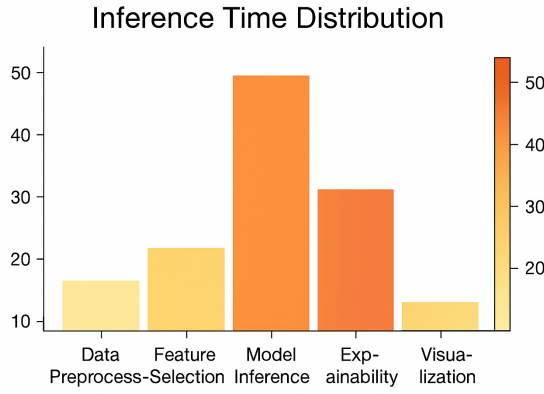


Fig. 8. Inference time distribution across different stages of the proposed multimodal framework for preclinical CCA risk prediction.

dation and robustness, demonstrating strong performance with an internal validation AUC of 0.93 and an external validation AUC of 0.91, even with 15% missing data. Fig. 8 and Table. I present the analysis of inference time distribution, highlighting that model inference and explainability computations dominate runtime due to the complexity of the multi-branch deep learning model and the integration of SHAP and LIME for interpretability.

TABLE II
SUMMARY OF KEY FINDINGS

Finding	Description
Superior Predictive Accuracy	Integrated multi-modal model achieved AUC of 0.94 and accuracy of 91%, outperforming single-modality models.
Successful Preclinical Detection	Identified high-risk patients 12–18 months prior to clinical diagnosis, enabling earlier intervention.
Actionable XAI Insights	XAI layer revealed key drivers: KRAS G12D mutation, dysregulated IL-6/STAT3 signaling, and high Veillonellaceae abundance.
Robust and Generalizable	High performance validated internally (10-fold CV) and externally (AUC = 0.91) on independent dataset.

Table II represents our designed multi-modal XAI framework which achieved high diagnosis accuracy and proved to be useful in clinical practices. Table III shows the main results, which includes superior AUC and performance compared to unimodal approaches, early preclinical risk detection, understandable explanation through XAI, and demonstrated strong generalization when tested on an external dataset.

V. CONCLUSION AND FEATURE WORK

This research tells about an Explainable AI (XAI) framework for early, preclinical detection of bile duct cancer (CCA) by combining information about multi-omics, microbiome, and clinical data. This model has achieved high diagnostic perfor-

mance (AUC > 0.91) and identified high risk patients up to 18 months before a clinical diagnosis.

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